MINERALS FROM THE SEA

For centuries, people have been using the oceans as a source of salt, and also collecting other materials ranging from beach sand to pearls. But the mineral resources of the seafloor were almost unknown until quite recently, when the technology needed to exploit them was developed. The most important of these resources are oil and natural gas, tapped from reserves buried deep in the rocks of the continental shelves. The seas are also a major source of gravel and sand, and even diamonds. But many oceanic minerals are too difficult and expensive to harvest, especially those found on the deep ocean floor.

▲ **SEA SALT**
The most abundant mineral in ocean water is sodium chloride, or common salt. It is extracted by evaporating seawater in shallow, sunlit salt pans. This simple process has been used for thousands of years, and it still meets roughly a third of the world's need for salt.

► **FRESH WATER**
Some countries use the sea as a source of water for drinking purposes and crop irrigation, but the salty seawater has to be pumped through a desalination plant to remove the salt. It is an expensive process that uses a lot of energy, so it is mainly used in the rich desert states, especially in the Middle East. The desalination plant shown here is in Kuwait, on the desert shores of the Persian Gulf, and nearly a quarter of the world's desalinated water is produced in neighbouring Saudi Arabia.

OIL AND GAS

The thick sediments covering many seafloors hold large reserves of oil and gas, formed from the decomposed remains of marine organisms. Initially, only the reserves beneath shallow continental shelves were exploited, but offshore platforms, like the one shown below, now work in water up to 10,000 ft (3,000 m) deep. They may drill 16,500 ft (5,000 m) or more below the seabed.